

Date.....

SEMI CONDUCTOR

The conductors lie b/w conductor and insulator the known as Semiconductor.

* Energy Band diagram metal and conductor :-

→ All metals are very good conductor of current. They allow very large flow of current to them.

→ Group-1, Group-2, Group-3 in periodic table elements are conductors.

→ metals are having metallic bonding.

	Conduction Band	$\rightarrow e^-$
	Valance Band	$E_g = 0$
		Energy gap
		for conductor

→ In metals free electron are available $t = 0$ K.

→ All metals are unipolar. (only having electron).

→ At temperature increases in valance band and conduction band will overlap each other.

→ A metals e^- concentration is very high. That is

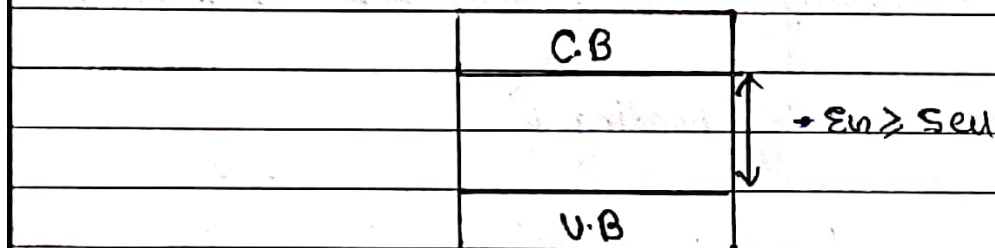
$$n = 10^{28} / m^3$$

→ A metals there will be only drift current.

→ Ex Gold, copper, zinc, platinum, silver, Iron etc.

* Energy band diagram of a insulator →

- (i) Insulator is a bad conductor. of current they do not allow any current flow of current through them
- (ii) Group - 7 Group 8 in periodic table known as insulator
- (iii) insulator have ionic bonding.
- (iv) The energy band gap of insulator is very large.

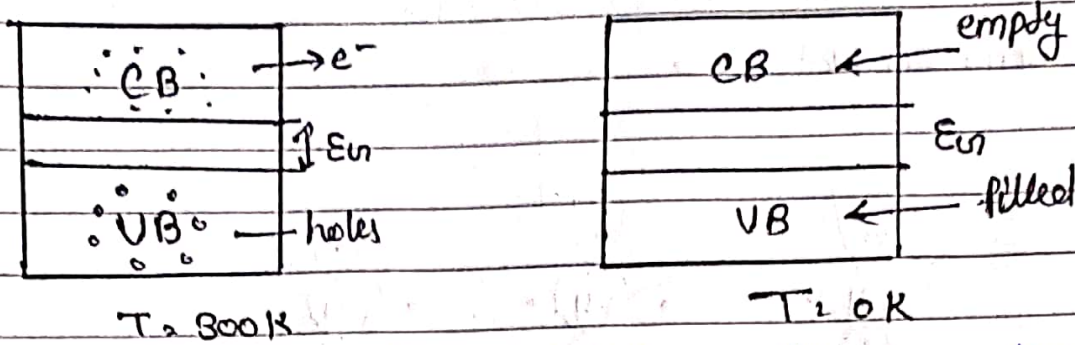


- (v) The conductivity of insulator is 0.
- (vi) in insulator slightly decreases with temp.
- (vii) Ex → air, SiO_2 , mica, wood, plastic, glass, diamond and rubber.

* Energy band diagram of semi conductor:

- (i) Semi conductor are the element whose conductivity lies b/w the conductivity of insulator or conductivity of conductor.
- (ii) Group - 4 in periodic table is known as semi conductor
- (iii) Semi conductor has covalent bonding.
- (iv) Semi conductor are bi-polar. That is having two different type of charge carrier electrons and holes.
- (v) Energy gap is small

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(vi) At 0 K semiconductor behaves like an insulator.

(vii) $E_g \downarrow$ Temp $\uparrow$

(viii) In semiconductor there is a diffusion current.

(ix) Ex $\rightarrow$ Silicon, Germanium, Gallium, As, C etc.

Ques Why carbon is not considered as a semiconductor element.

Ans Carbon belongs to fourth group of the periodic table but it is never considered as a semiconductor element due to

(i) larger energy gap (greater than 1.5 eV)

(ii) Unstable properties.

(iii) Unreliable properties.

(iv) Unpredictable properties.

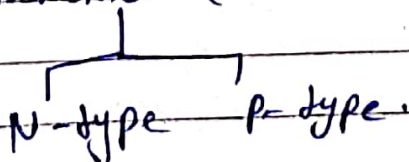
(v) Carbon some time working as graphite and some time working as insulator.

Ques Explain type of semiconductor with crystalline structure at $t = 0\text{ K}$ and $t = 300\text{ K}$ $t \rightarrow \text{temp}$.

Ans There are two types of semiconductor

(i) intrinsic (pure s/c)

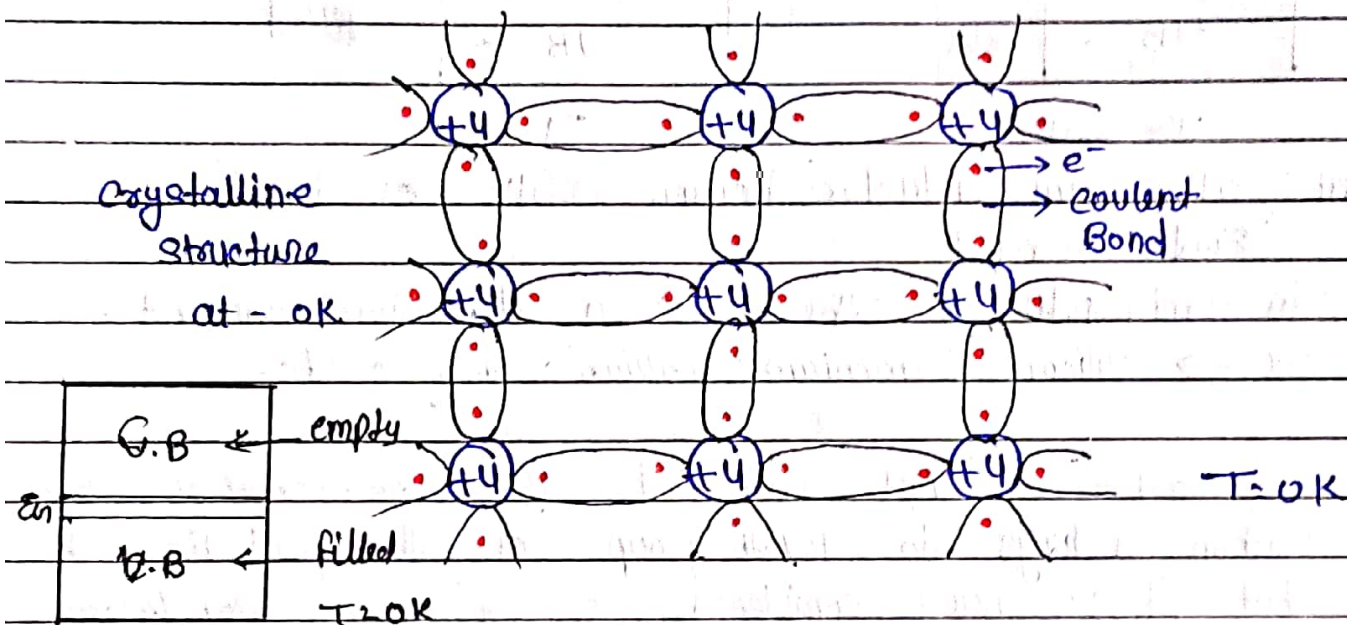
(ii) extrinsic (lab)



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* Intrinsic Semiconductor $\rightarrow$ 

$\rightarrow$ Intrinsic sic is a pure semiconductor. It is a known the ~~dehenerated~~ semiconductor. The sharing of electron with ~~neabing~~ atom ~~neabing~~ atom is called covalent bond. In one covalent bond there will be to balance electron at 0K all balance electron are perfect covalent bonding. ~~Intern~~ Intrinsic sic at $T=0\text{K}$ will be working as Insulator.

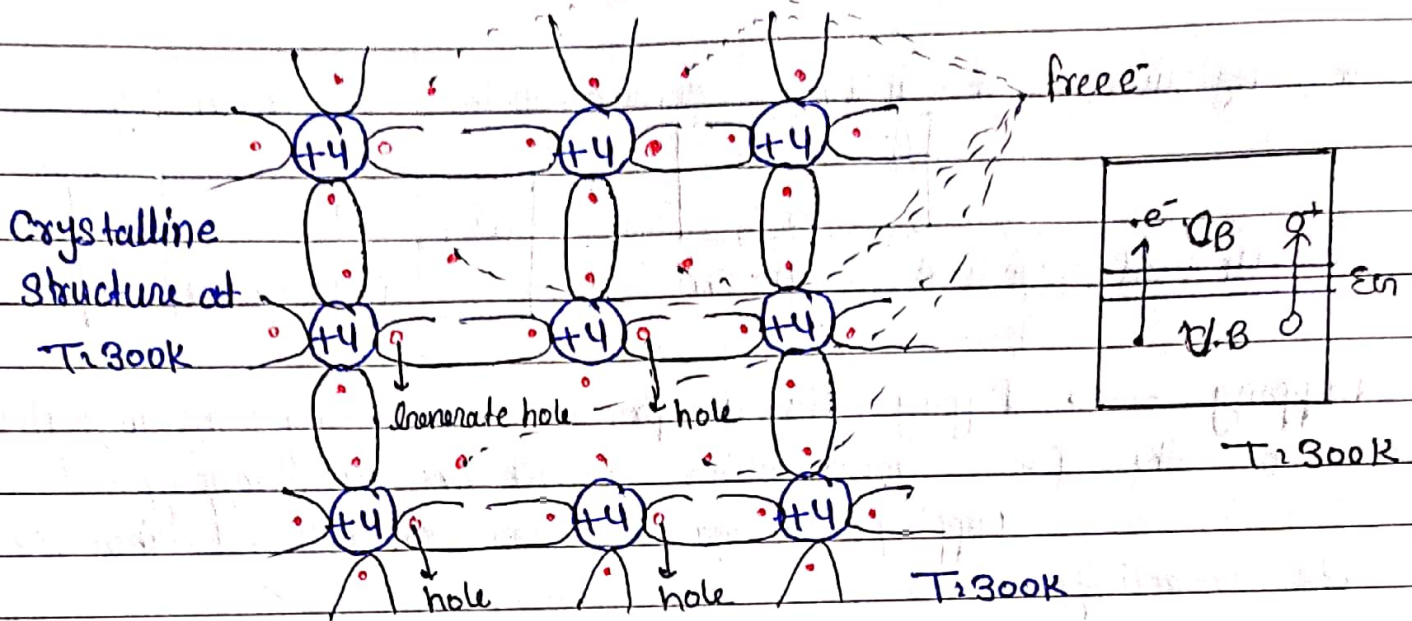
At 300K

> When a covalent bond is broken it will create electron and hole.

> The electron will be jumping V.B to C.B and become free electron. and hole will be remain in V.B.

> The hole is define deficiency of a electron in a broken covalent bond.

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→ Hole is a carrier of current with a charge of $+1.6 \times 10^{-19} C$.

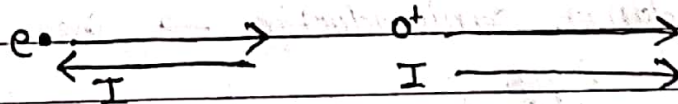
→ at $T = 300K$ The concentration of electron equals to concn of holes.

$$n = p$$

→ The condition of intrinsic semiconductor -

$$n_i = n = p$$

→ The generated current I is flow opposite direction of charge carriers of electron. and generated current of holes flows in the same direction.



→ The conductivity of intrinsic semiconductor is given by

$$\sigma_i = n_i q (u_n + u_p)$$

Conductivity of Intrinsic semiconductor Intrinsic charge carrier mobility of e^- mobility of holes.

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→ The resistivity of intrinsic semiconductor is given by

$$\rho_i = \frac{1}{\sigma_i}$$

Unit of resistivity, $\Omega\text{-cm}$

$$\rho_i = \frac{1}{n_i q (\mu_n + \mu_p)}$$

Doping → Doping is a process of adding the impurities of the pure semiconductor is known as doping. Doping increase carrier concn and increase the conductivity.

There are two types of doping.

i) Trivalent / Acceptor impurities → B, Al, Ga, In

ii) Pentavalent / donor impurities → P, As, Sb, Bi

Lightly doped
 $1:10^{11}$

Heavily doped
 $1:10^5$

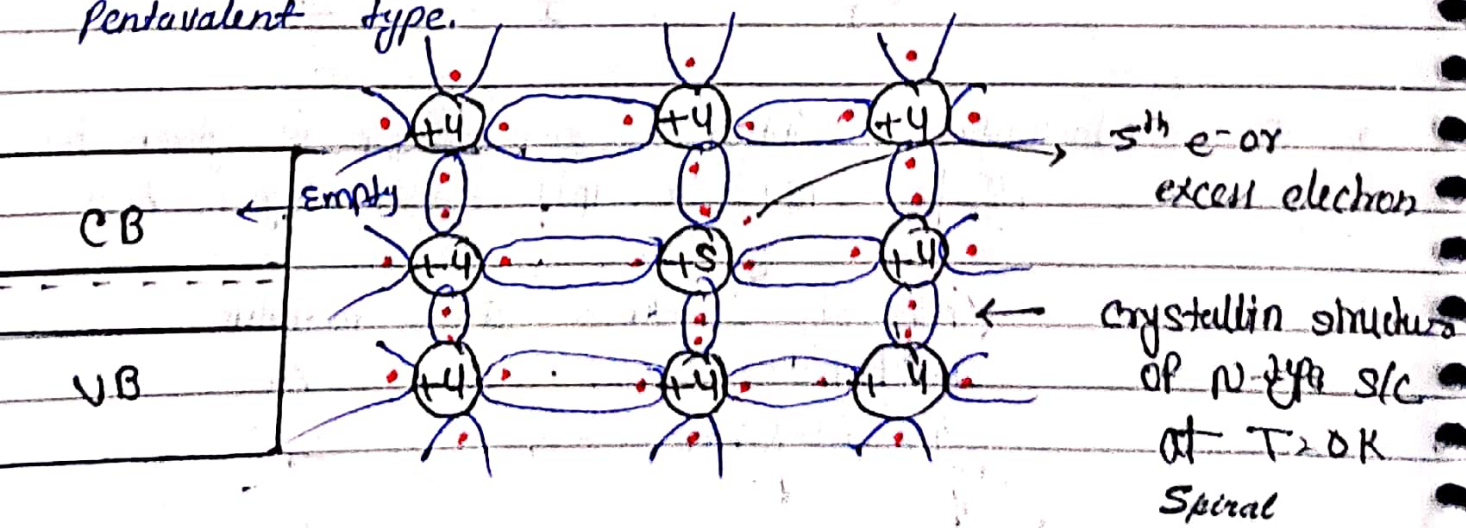
moderately doped
 $1:10^4 \text{ to } 10^5$

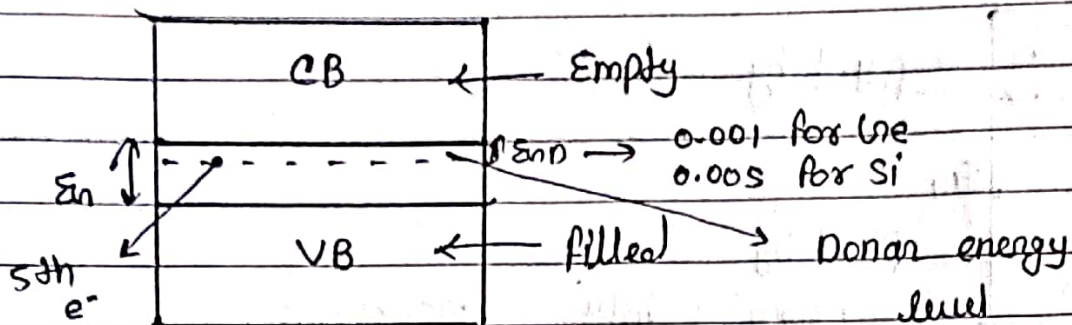
Statement → The minimum doping required to convert intrinsic semiconductor to extrinsic semiconductor is $1:10^8$.

Extrinsic semiconductor →

There are two types of extrinsic semiconductor

N-type / donor semiconductor. → The impurities in added pentavalent type.





Energy gap is independent of temp.

→ Donor energy level is a discrete energy level of all the pentavalent atoms added to the semiconductor. Created

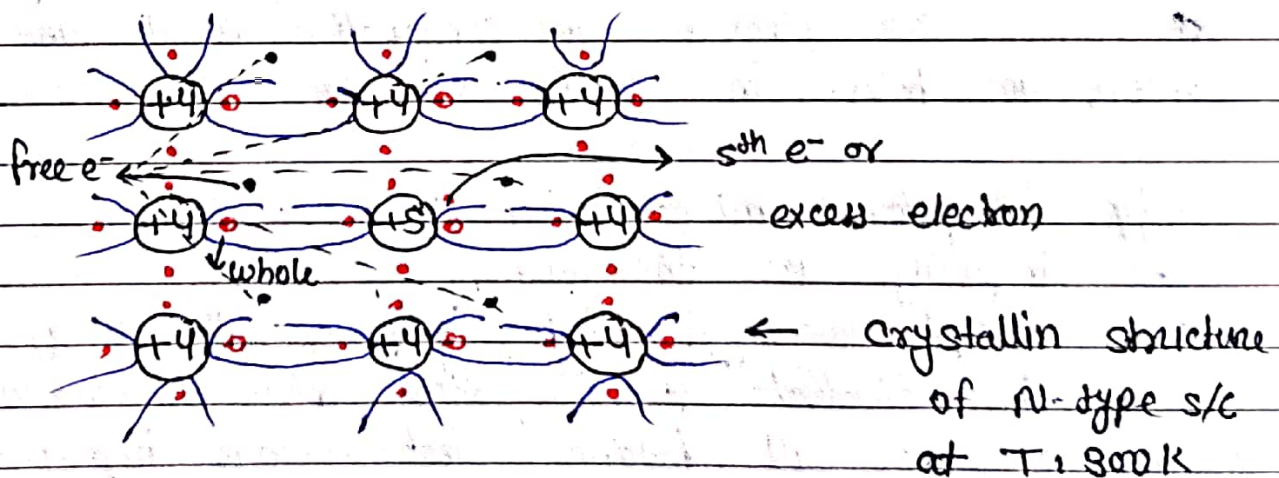
→ just below the conduction band.

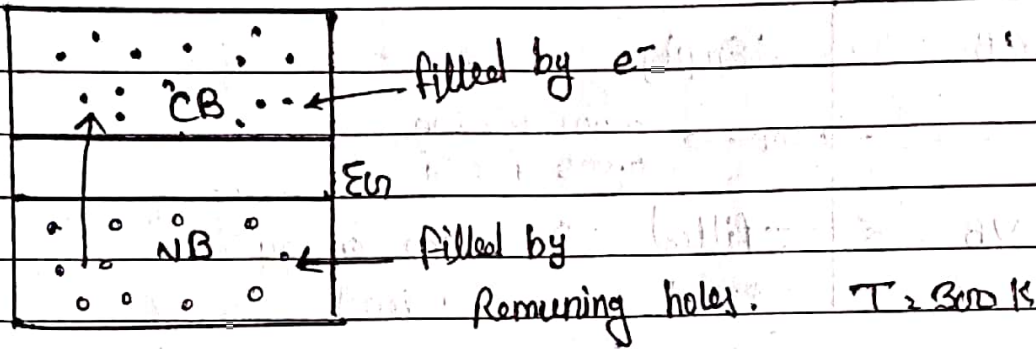
→ Donor energy level denotes energy level of all the pentavalent atoms added to the semiconductor.

→ At 0 K the fifth electron of all the impurity atoms are existing at the donor energy level.

→ The fifth electron is not in covalent bonding and it is not a free electron. It is known as bounded electron.

→ n-type semiconductor at $T = 0\text{ K}$ working as insulator.





- In entire semiconductor every impurities will be donating one electron in conduction band and it is also known as donors.
- The process where the fifth electron are moving from ^{donor} energy level into C.B. is called donor level ionization.
- Donor level ionization increase with the temp.
- Above 300 K there is no donor ionization.
- The majority carrier in conduction band will contribute more current with less noise and minority carrier in CB contribute less current with more noise.

Statement → In n-type semiconductor the majority carriers are electrons in conduction band and minority carriers are holes in conduction band.

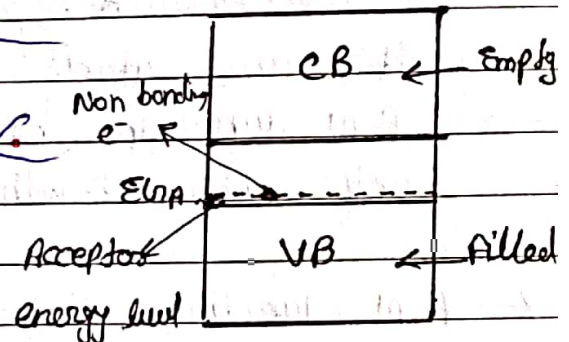
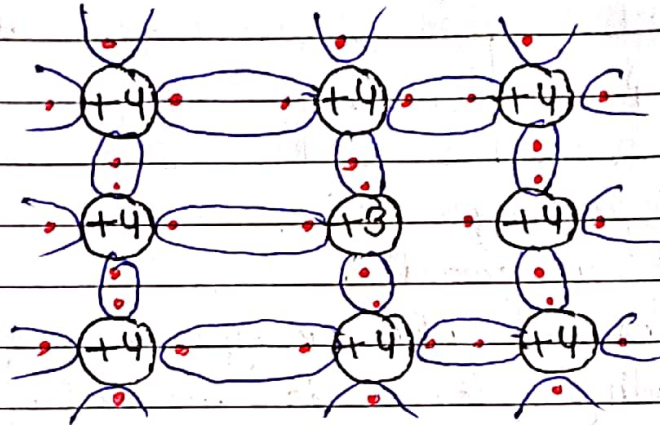
* P-type / Acceptor semiconductor :-

The impurity is trivalent.

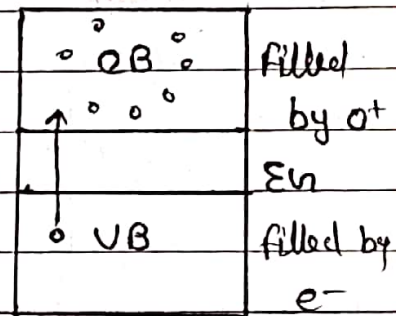
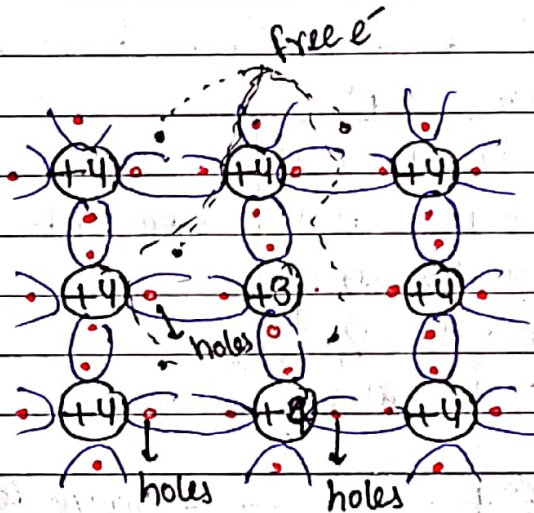
- The acceptor energy level indicates the energy level of all the trivalent atoms added the semiconductor.
- At $T = 0K$ all balance electron are imperfect covalent bonding.

→ p-type semiconductor at $T=0K$ work as insulator.

Crystalline
Structure at
 $T=0K$



Crystalline
Structure
at $T=300K$



$T=300K$

$T=300K$

→ In p-type semiconductor every impurity atom will be receiving one electron to complete covalent bonding that's why p-type known as acceptor.

→ In p-type semiconductor the concn of hole is increases over the electrons concn.

→ The majority carriers in p-type semiconductor are holes concn and minority carriers are electron concn.

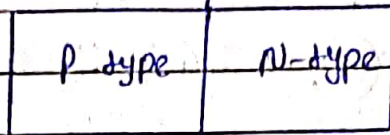
→ In p-type semiconductor the current dominating by holes.

→ The conductivity due to minority carrier is almost negligible.

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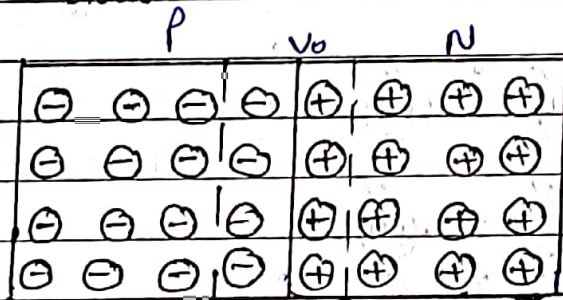
P-N Junction →



- The rectification property is exact.
- It can work as rectifier.
- P-n junction created b/w P-type & n-type semi conductor only when bonding force is created.

P-N Junction Diode.

Imp In
Exam



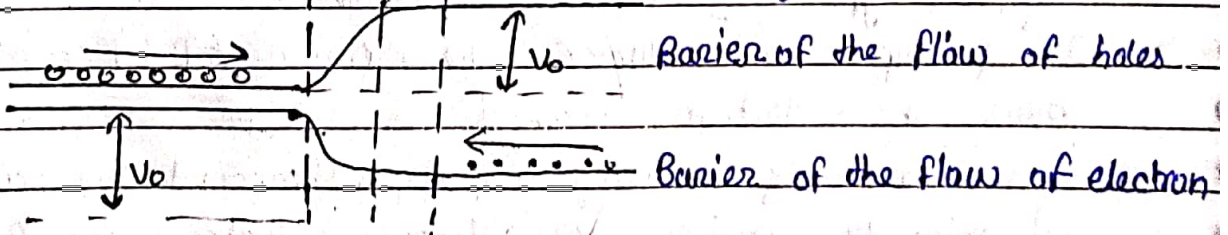
Depletion layer.

Charge density

Electric field

$$E = -\frac{dv}{dw}$$

$$|E| = \frac{1}{w}$$



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Depletion layer is also called space charge region.

Depletion layer is created due to majority carriers across the junction.

Depletion layer consists of a large no. of ions and covalent bond.

Depletion layer consists negative charge and positive charge side of the junction.

In the depletion region mobile charge carriers are zero.

Depletion layer opposes majority carriers in crossing the junction.

i) Depletion layer will not oppose the minority carriers crossing the junction.

iii) Depletion layer will help the minority carriers crossing the junction.

i) V_0 is called potential hill, diffusion voltage, Barrier Voltage, Built-in Voltage, Contact potential.

ii) For Ge diode or for Si diode.

Ge

Si

$V_0 = 0.1V$ to $0.5V$

$V_0 = 0.6$ to 0.9

typical value ≈ 0.2

typical value ≈ 0.7

xi) The depletion layer of the diode can not be measured practically by using a volt meter.

xii) Diode can not work as battery.

xiii) The breadth of depletion

$w \propto \sqrt{N_A}$

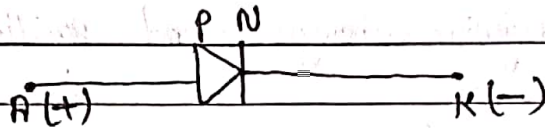
doping

xiv) In p-n junction diode p and n regions are having equal doping concentration.

xv) If any p-n junction field intensity is always maximum at junction.

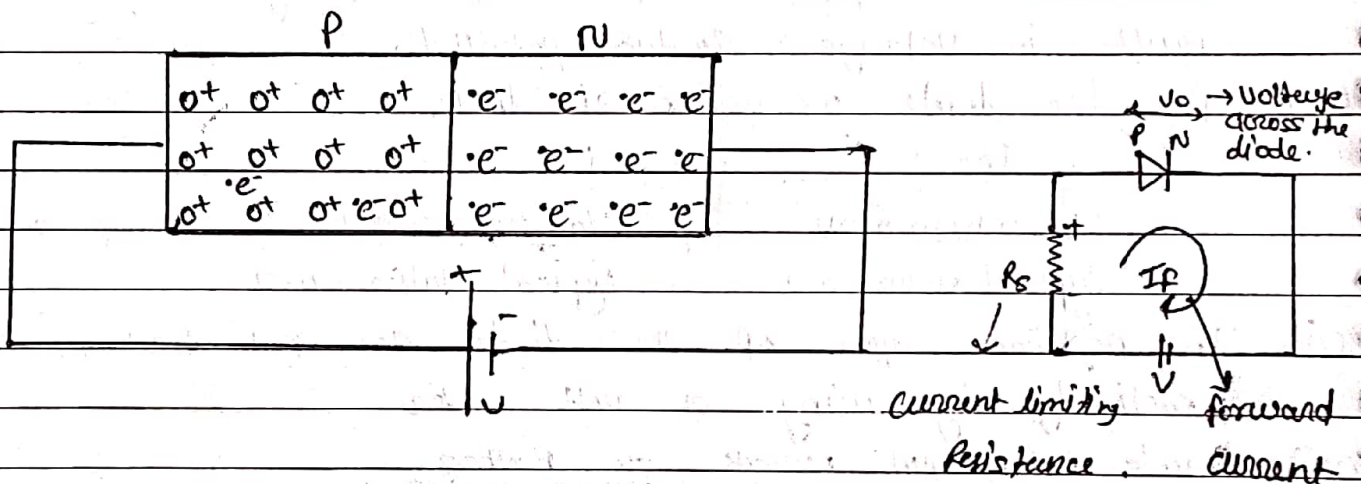
Along with temp. a proper biasing must be applied so that majority carrier can overcome barrier potential and they will contribute the current.

= Symbol of diode $\rightarrow$



The arrow mark symbol denote the direction of forward current or direction of conventional current. when diode is forward bias.

P-N Junction diode in forward bias:-



When P-N junction is forward bias.

The breadth of depletion layer is reduces.

The barrier height can not oppose the majority carriers in crossing the junction. so that the barrier effect is nullified.

The forward current is due to only majority carriers.

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$$I_f = I_s \left[e^{\frac{V_f}{nV_T}} - 1 \right]$$

$$I_f \approx I_s e^{\frac{V_f}{nV_T}}$$

Here V_T = thermal voltage

= 26 milli volt for room temp.

n = Recombination factor

Si $\Rightarrow n = 2$

Ge $\Rightarrow n = 1$

I_s = saturation current or Reverse saturation current

V_d = forward voltage

* Cut in voltage $\rightarrow$ It is also known as offset voltage, threshold voltage, knee voltage, break voltage.

$\rightarrow$ Cut in voltage is defined as the minimum forward voltage required across the diode so that current will pass into the diode.

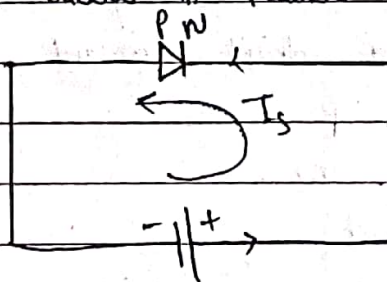
for Ge = 0.1 to 0.5V

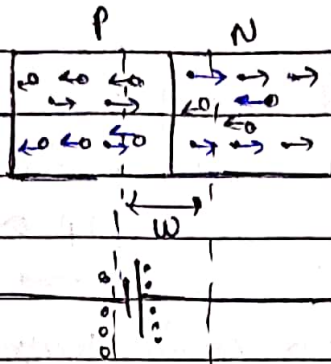
typical value = 0.2V

for Si = 0.6 to 0.9V

typical value = 0.7V

* P-n junction diode in Reverse bias.





- 1) The breadth of depletion layer is increases.
- 2) The barrier height increases.
- 3) In reverse bias diode current only due to minority carriers and current is represented I_s or I_0 .
- 4) I_0 is called reverse saturated current or leakage current or minority current or thermally generated current.
- 5) I_0 of Ge diode $>$ I_0 of Si diode.
- 6) I_0 is highly sensitive to the temp.
- 7) I_0 double for every 10°C .

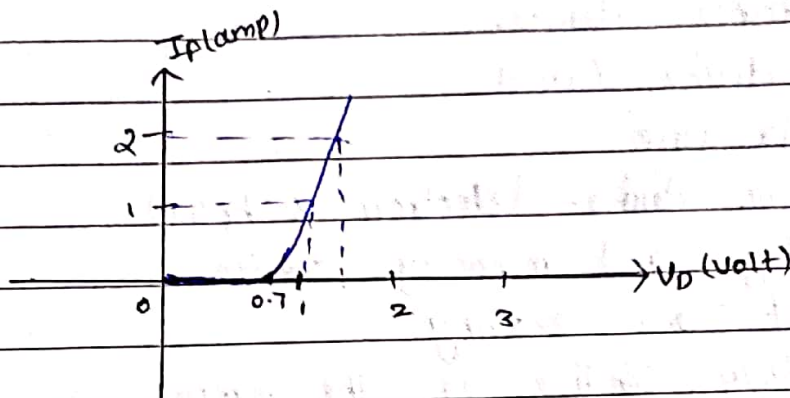
$$I_0(T_2) = I_0(T_1) \left[2 e^{\left(\frac{T_2 - T_1}{10} \right)} \right]$$

- 8) In reverse bias diode the majority carrier of p-n region will move away from the junction and they will accumulate and store in the device. They didn't contribute any type of current in p-n junction diode.
- 9) Since majority carrier are blocked in crossing the junction so reverse bias is also called blocking bias.
- 10) reverse current I_0 is a drift current.
- 11) I_0 flow from n to p.
- 12) I_0 flow from p to n.
- 13) I_0 is called reverse current.

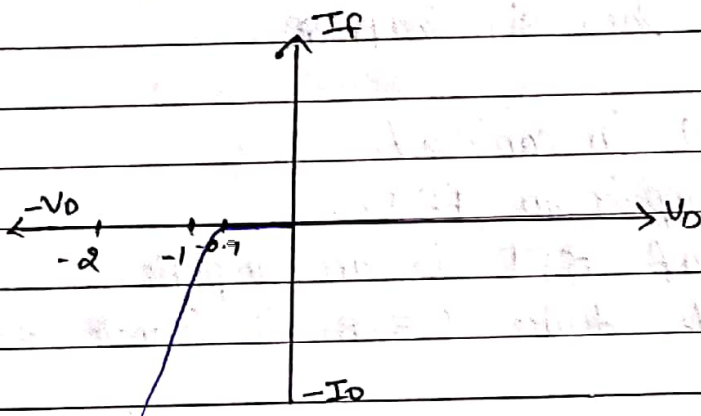
$$I_0 = -I_s \left[e^{-\frac{V_0}{nV_T}} - 1 \right]$$

* V-I characteristics of p-n Junction diode :-

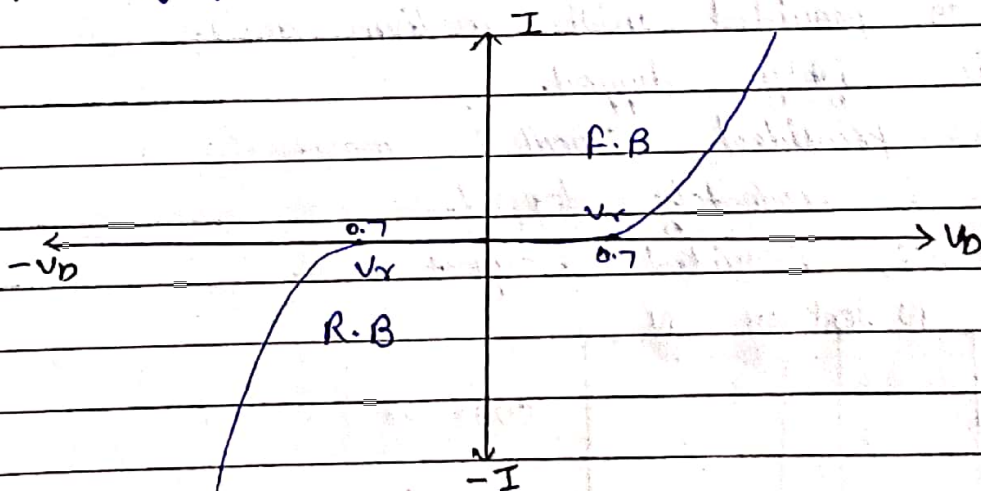
① when p-n junction is in forward bias.



② when p-n junction is in Reverse bias.

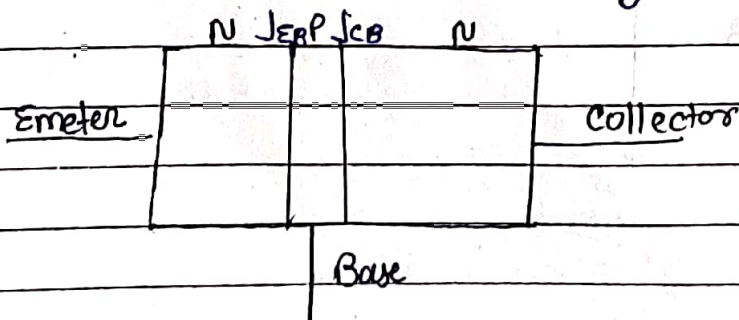


* Complete graph



* Bipolar Junction Transistor (BJT)

- i) Invented by William Shockley, Bardeen, Barteene
- ii) low input resistance device.
- iii) Current Control device (CCD)
- iv) Power dissipation is more
- v) It is a bipolar device. (electron & holes)
- vi) It has majority and minority carrier
- vii) leakage current are exciting.
- viii) It is a highly sensitive to the temp.
- ix) Thermal stability is less. compare to FET
- x) It can be fabricated Ge, Si, GaAs.
- xi) Cut-in voltage exist.
- xii) Gain Band width (GBW) is constant.
- xiii) It has large gain compare to FET.
- xiv) The major Application of BJT is an amplifier.
- xv) It has three terminals device (E, B, C) (Emitter, Base, Collector)
- xvi) Emitter is highly doped to inject majority carrier into the base
- xvii) Emitter is provided with medium area.
- xviii) Base is lightly doped.
- xix) Base is provided smallest area.
- xx) Collector is moderately doped.
- xxi) collector is provided largest area.



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* Modes of transistor :-

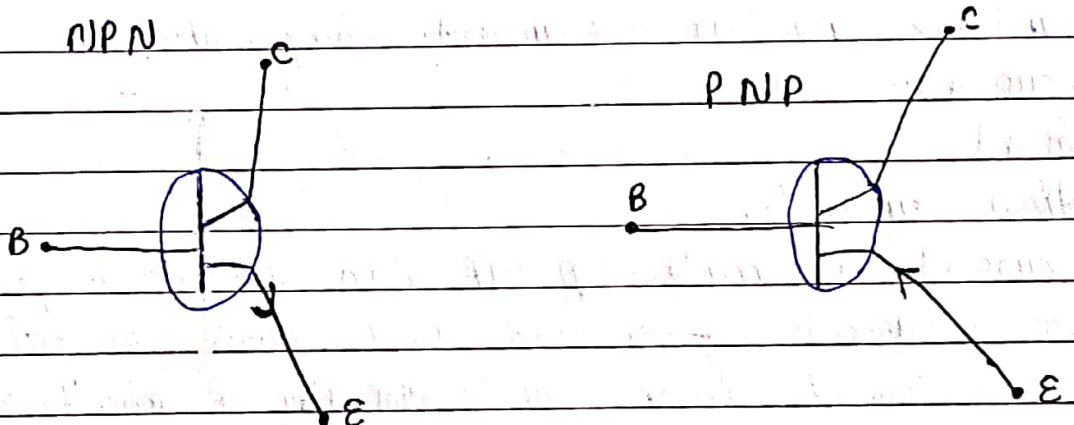
	I_E	I_C	Tr mode
I_E	R.B	R.B	cutoff (switch)
I_C	F.B	F.B	saturated
	F.B	R.B	Active
	R.B	F.B	Reverse Active.

* Types of BJT

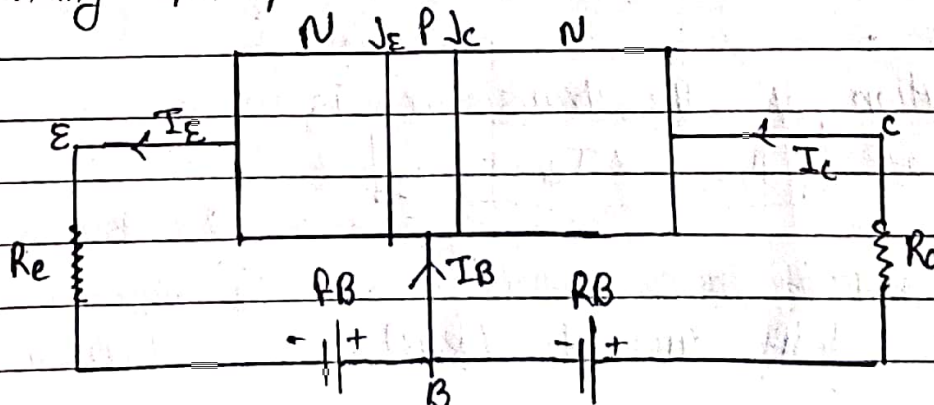
There are two types of BJT

- N-P-N transistor
- P-N-P transistor

* Symbolic representation of BJT



* Working principle of N-P-N transistor



Apply KCL

$$I_E = I_B + I_C$$

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- 1) In n-p-n transistor the recombination occurs in the base region and emitter region.
- 2) In p-n-p transistor the recombination occurs base region and collector region.
- 3) In a transistor internal currents are bipolar. (due to e^- and holes)
- 1) Emitter current is a forward current.
- 2) Emitter current is a majority current.
- 3) Emitter current is a diffusion current.
- 4) In n-p-n transistor base current is due to holes.
- 5) In p-n-p transistor the base current is due to electrons.
- 6) Collector current is a diffusion current.
- 7) The recombination current flows in BJT.

In reverse bias region the conduction of current is also known as I_{EB} I_{CO} .

I_{CO} is a drift current.

In a transistor are varies current component

diffusion current

drift current

recombination current.

Collector current is made up off two current component

majority carrier current $\rightarrow$ 95 to 96 % emitter current

minority carrier current $\rightarrow$ due to the flow of minority carrier in reverse bias region.

A General equation of the transistor is

Collector current of $I_C = \alpha I_E + I_{CO}$

the transistor.

$\alpha = 95\% \text{ to } 96\%$

I_{CO} : minority carrier current

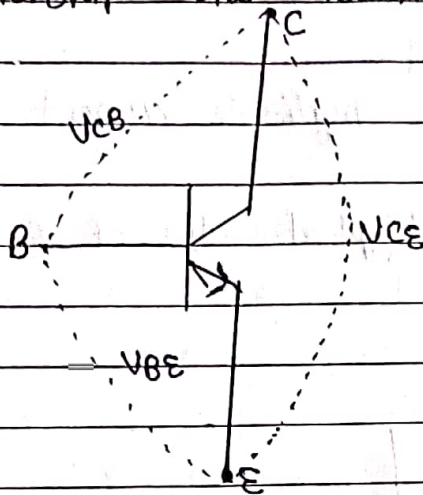
: drift current (R.B)

I_E : Majority carrier current

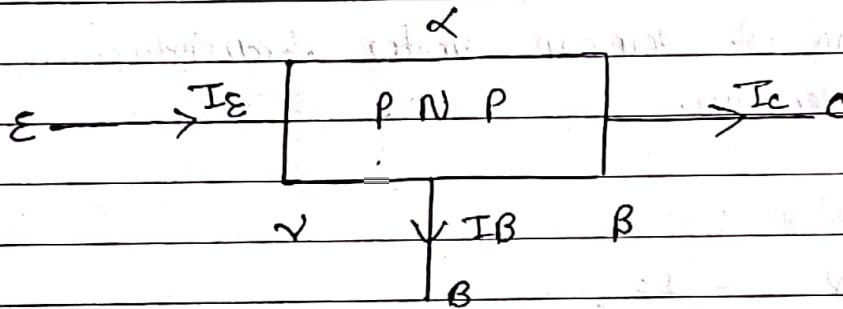
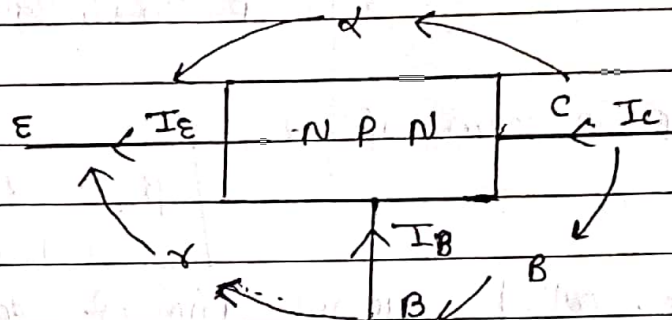
diffusion current

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* Relationship b/w terminal voltages of transistor $\rightarrow$



$$V_{CE} = V_{BE} + V_{CB}$$



$$I_E = I_B + I_C$$

$$I_E = -V_E$$

$$I_B \text{ or } I_C = +V_E$$

* Alpha $[\alpha]$ of the transistor :-

$$\alpha = \frac{-I_C}{I_E}$$

Since I_C & I_E have opposite side

$$\alpha = \frac{I_C}{I_E}$$

α will be positive.

α is slightly less than 1

$$\alpha \approx 1$$

In ideal transistor $\alpha = 1$

α is a current gain of common base transistor.

$$\alpha = \frac{\beta}{1 + \beta}$$

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ii) Beta [β] of the transistor :-

$$\beta = \frac{I_c}{I_B} \quad \beta = \frac{I_c}{I_B} \text{ is negligible current}$$

$$I_c \gg 1$$

$$\beta \gg 1 \quad (\text{typical value of } \beta, 49)$$

$$\boxed{\beta = 30 \text{ to } 300}$$

β in term of α :-

$$\boxed{\beta = \frac{\alpha}{1-\alpha}}$$

β is called current gain of common emitter transistor.

β is a temperature sensitive.

i) Alpha (α) of the transistor :-

$$\alpha = - \frac{I_E}{I_B}$$

$$\alpha = \left| \frac{I_E}{I_B} \right| \quad \alpha = +ve$$

$$\alpha = \left| \frac{I_B + I_C}{I_B} \right| \quad \text{due to KCL}$$

$$\boxed{\alpha = 1 + \beta}$$

$$\alpha = 1 + 49$$

$$\boxed{\alpha = 50}$$

typical value.

Relationship between α , β , γ

$$\gamma = 1 + \beta = 1 + \frac{\alpha}{1-\alpha} = \frac{1-\alpha + \alpha}{1-\alpha} = \frac{1}{1-\alpha}$$

$$\boxed{\gamma = 1 + \beta = \frac{1}{1-\alpha}}$$

Spiral



→ In a transistor various current gains α , β , γ arranging them is the ascending order.

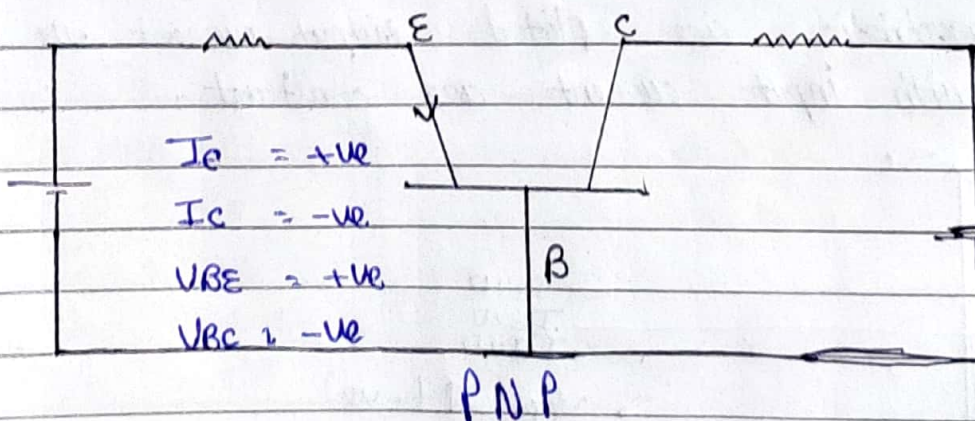
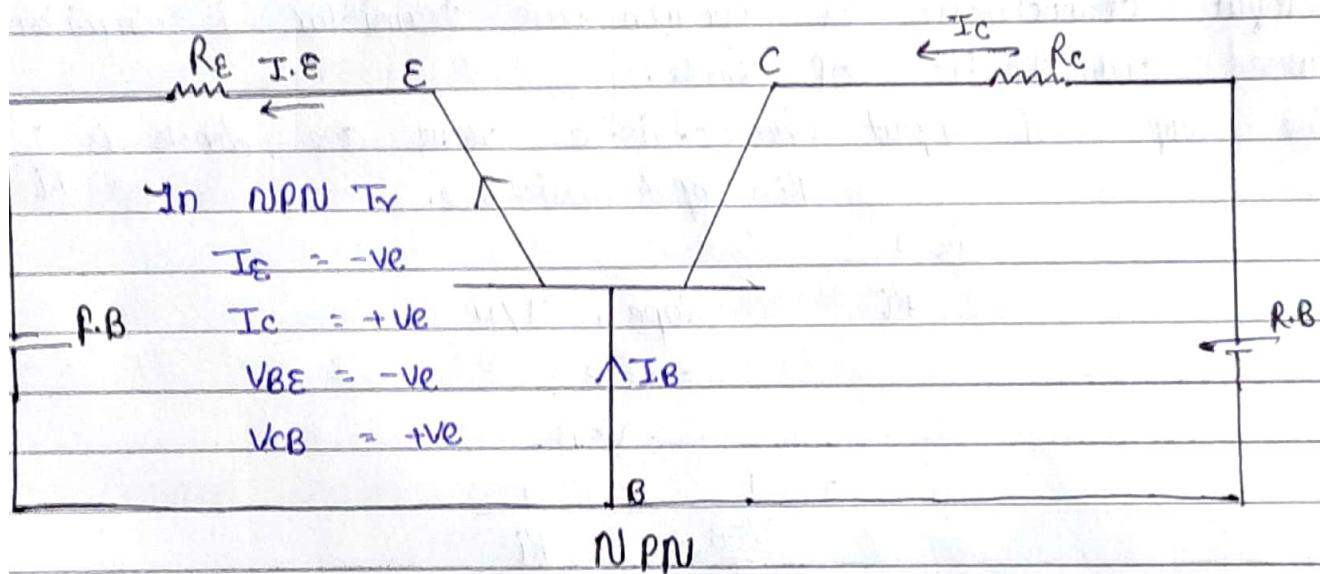
$$\alpha < \beta < \gamma$$

$$\alpha > \beta > \gamma$$

Characteristics of the transistor

There are three types of characteristics of transistor

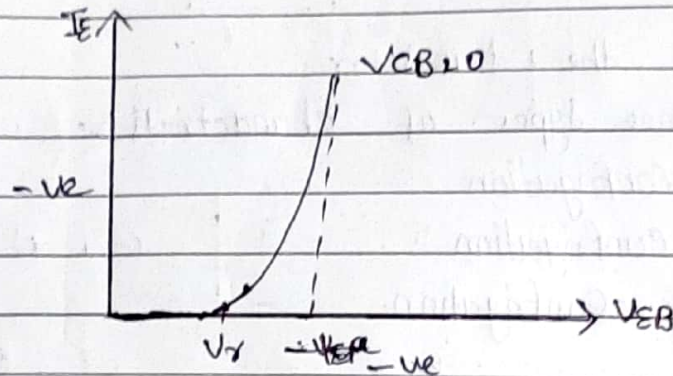
- Common base Configuration
1. Common emitter Configuration
3. Common collector Configuration.



Input characteristics of the Common base :-

→ Input characteristic are plotted input current I_E vs input voltage V_{EB} by keeping output voltage as constant.

I_E vs V_{EB}



The input characteristic of common base transistor is similar to forward characteristic of diode.

→ The slope of input characteristic curve denote is $\frac{1}{R_i}$
 R_i : input resistance.

is $\frac{1}{R_i}$

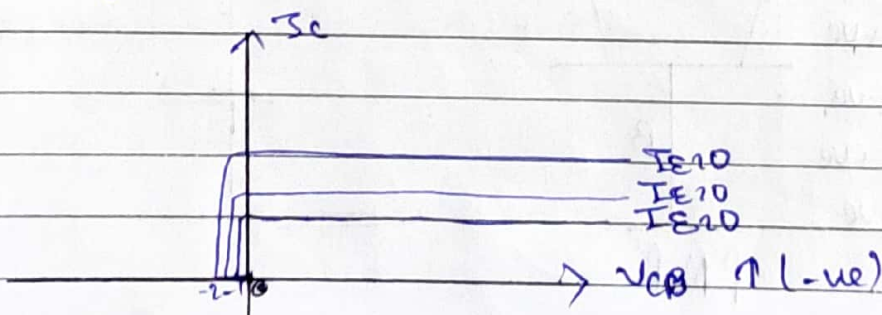
Slope, $\frac{Y}{X}$

$$= \frac{I_E}{V_{EB}}$$

$$= \frac{1}{\frac{V_{EB}}{I_E}} = \frac{1}{R_i}$$

Output characteristics of Common base :-

→ output characteristic are plotted output current I_C vs output voltage with input current are constant.



The slope of

$$\text{O/P characteristic} = \frac{1}{R_o}$$

- CB transistor can work as constant current source.
- CB transistor is current control current source device.
- The slope of CB transistor is $1/R_o$
- The slope of out

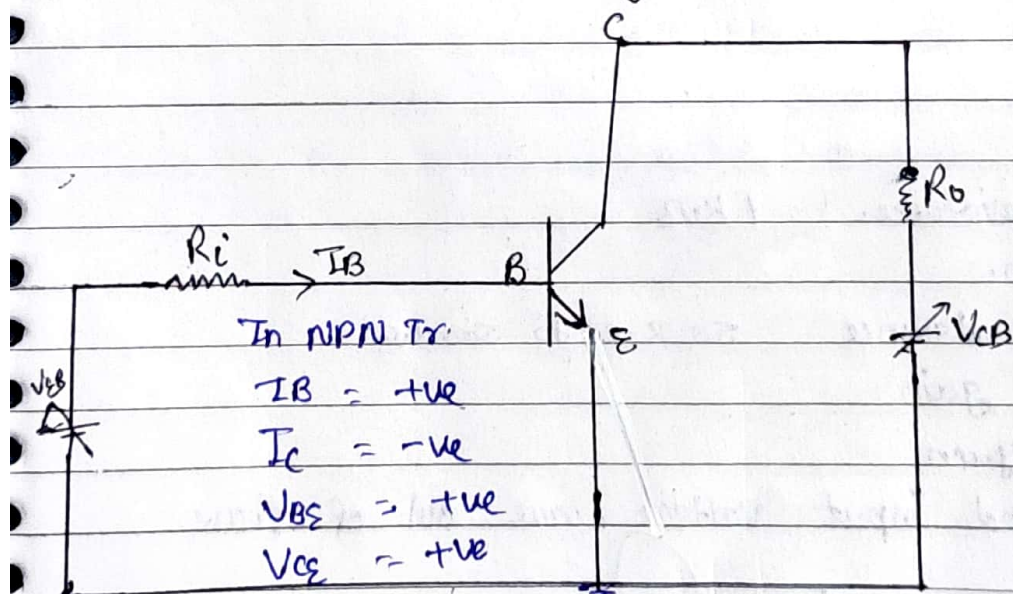
* Properties :-

- Lowest input resistance. $R_i \leq 100 \Omega$
- Highest output resistance $R_o > 1 M\Omega$
- lowest current gain $A_i = \left| \frac{I_c}{I_e} \right| = \alpha \approx 1$

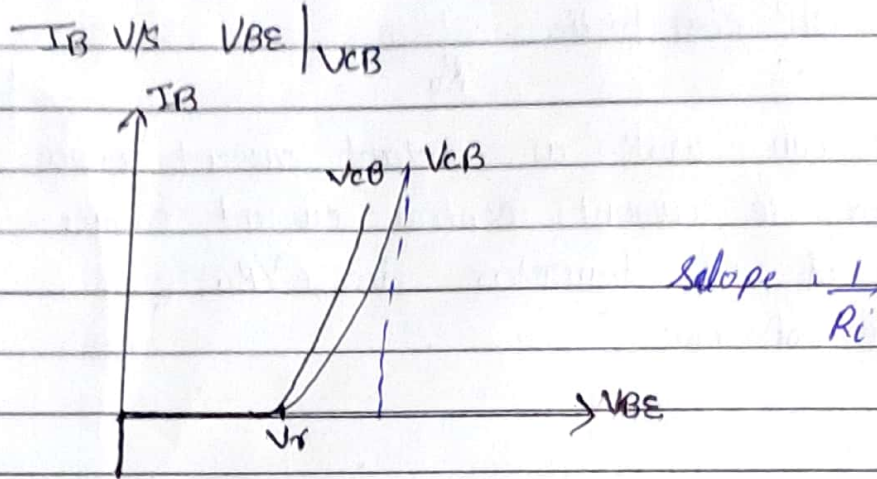
* Application →

- As a known inverting voltage amplifier.
- As a ~~current~~ constant current source.
- It is used to high frequency amplifier.

* Common Emitter Configuration →

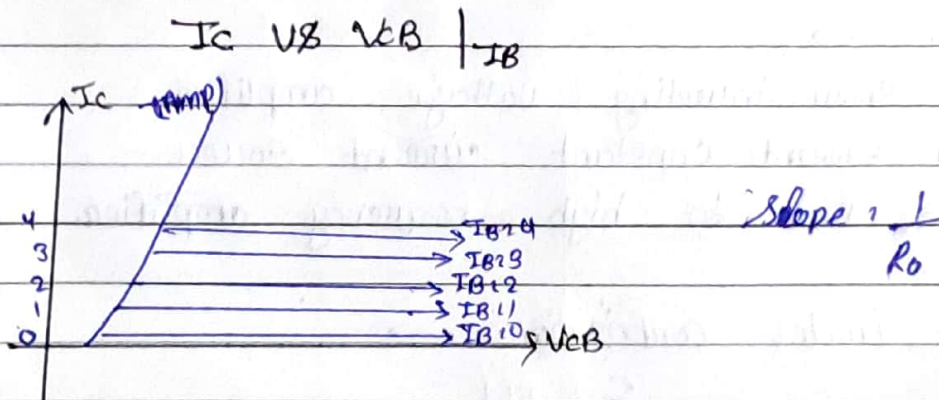


← Input characteristic of common emitter :-



→ Input characteristic of c.e transistor is similar to forward characteristic of diode.

* output characteristic of common emitter :-



- Properties →

- moderate input resistance. 1 KΩ
- Highest Power gain.
- moderate output resistance 50 KΩ to 500 KΩ
- moderate current gain
- moderate voltage gain
- output voltage and input voltage are out of phase.



* Advantage :-

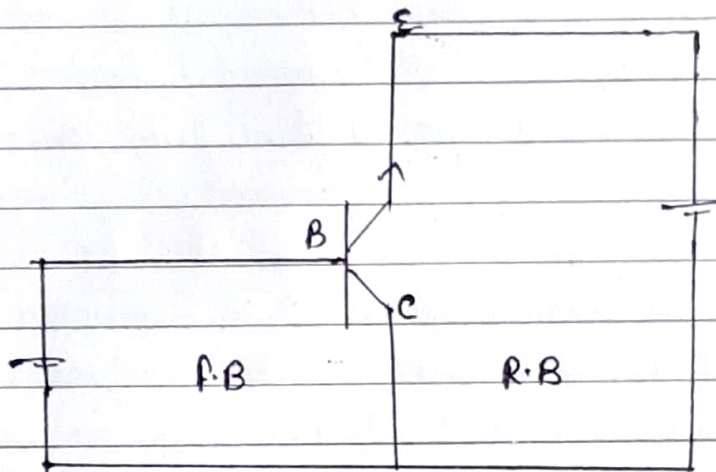
There are two types of advantage

- (i) Highest power gain
- (ii) 180° phase shift

* disadvantage :-

→ Input resistance is very very small.

* Common collector configuration →



→ It is very popularly known as emitter follower ckt

→ In plotting equivalent circuit all biasing voltages are short circuited their by collector region is grounded hence the circuit is called common collector circuit.